SYSTEM DESIGN DOCUMENT
LITPATH AI: SMART PATHFINDER FOR THESES AND DISSERTATIONS

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INTRODUCTION

This System Design Document provides a comprehensive description of the system architecture and design for LitPath AI: Smart Pathfinder for Theses and Dissertations. The system is designed to enhance research discovery within the DOST-STII Library by addressing the limitations of the existing Online Public Access Catalog (OPAC), particularly in terms of search relevance, performance, and usability.

LitPath AI is a standalone, AI-powered web application that complements the existing OPAC system. It introduces semantic search, automated citation generation, bookmarking, and analytics to improve the accessibility and usability of theses and dissertations.

PURPOSE

The purpose of this document is to define how LitPath AI will be constructed from a technical and architectural perspective. It ensures that the system design aligns with the requirements specified in the Software Requirements Specification (SRS) and the goals outlined in the Business Case.

This document describes the system's architecture, components, database structure, hardware and software design, and security mechanisms. It serves as a reference for developers, stakeholders, and evaluators to understand how the system will be implemented.

SYSTEM OVERVIEW

LitPath AI is a web-based application designed to improve the discovery of academic research materials, specifically theses and dissertations stored in the DOST-STII Library.

The system leverages AI-powered semantic search using Retrieval-Augmented Generation (RAG) and Gemini AI to interpret user queries and return contextually relevant results. Unlike traditional keyword-based systems, LitPath AI understands user intent, enabling more accurate and efficient research.

Key functionalities include:

- AI-powered semantic search with ranked results
- AI-generated overview summaries with citations
- Advanced filtering by subject and publication date
- Automated citation generation (APA, MLA, Chicago, IEEE)
- Bookmarking and research history tracking

- User feedback collection
- Administrative analytics dashboard

Design Constraints

The design and implementation of LitPath AI are subject to several constraints:

- The system depends on external AI services (Gemini AI), which introduces limitations such as API latency and rate limits.
- Development is limited to a four-member team with no external funding for development resources.
- Full-text access to theses and dissertations is out of scope for the public; only metadata and abstracts are used.
- The system must be deployed within the project timeline ending June 2026.
- The system must comply with institutional security and data privacy policies.

Roles and Responsibilities

The development and design of LitPath AI involve multiple roles:

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Tracie Tomon	Project Manager	09922492939	tctomon@student.apc.edu.ph
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Project References

The system design is based on the following references:

- Software Requirements Specification (SRS) for LitPath AI v2.0
- LitPath AI Business Case Document

System Architecture

LitPath AI follows a three-tier architecture:

1. Presentation Layer (Frontend)
2. Application Layer (Backend & AI Processing)
3. Data Layer (Databases and External Systems)

Hardware:
The system is currently deployed in a local Proxmox-based server environment with the following specifications:

Processor:
Intel Core i5-6500 @ 3.20GHz (4 cores, x86_64 architecture)

Memory:
32 GB DDR4 RAM

Storage:
- 512 GB SSD (primary system and application storage)
- 500 GB HDD (backup storage)
- Total: 1 TB

Performance Benchmarks:
- Buffered Reads: 515.55 MB/sec
- Average Seek Time: 0.06 ms
- FSYNCS/sec: 583.03
- Regex Processing: 4,154,772 ops/sec

Network:
Ethernet (Intel I219-LM)

System Unit:
HP ProDesk 600 G2 MT

This hardware supports development, testing, and temporary deployment. Production deployment will transition to a cloud-based environment managed by DOST-STII.

Software:
The system is composed of modular components:

Frontend
- React.js
- Tailwind CSS
- Vite (build tool)

Backend
- Django (Python-based web framework)

- RESTful API architecture

AI and Search Components
- Gemini AI API (for summarization and query understanding)
- SentenceTransformers (embedding generation)
- ChromaDB (vector database for semantic search)
- Retrieval-Augmented Generation (RAG) pipeline

Database Systems
- Supabase (PostgreSQL) for relational data
- ChromaDB for vector embeddings

Key Modules
- User Authentication Module
- Search and Retrieval Module
- AI Analysis Module
- Bookmarking Module
- Citation Generator Module
- Feedback Module
- Admin Analytics Dashboard

DATABASE DESIGN

The system uses hybrid database architecture:

Relational Database (Supabase PostgreSQL) stores:
- User accounts
- Sessions
- Bookmarks
- Search history
- Feedback data
- Logs

Vector Database (ChromaDB) stores:
- Document embeddings
- Semantic representations of theses

Data Structure Characteristics
- Metadata includes title, author, abstract, year, subject

- Indexed for fast retrieval and filtering
- Supports semantic similarity search

The system ensures consistency between frontend modules and database fields to support reporting, filtering, and analytics.

Hardware and Software Detailed Design

Hardware:
The system leverages existing hardware infrastructure for development and testing. No additional hardware is required. The system is optimized for SSD-based performance and sufficient RAM allocation to support AI processing workloads.

Software:
Search Module
- Accepts natural language queries
- Converts queries into embeddings
- Retrieves relevant documents via vector similarity
- Reranks results using AI

AI Overview Module
- Uses Gemini AI to generate summaries
- Produces contextual explanations with references

User Interaction Modules
- Bookmarking system for saving research
- Citation generator supporting multiple formats
- Search history tracking

Admin Module
- Displays analytics on search trends
- Provides insights into user behavior

System Security and Integrity Controls
The system incorporates multiple layers of security:

Authentication and Authorization
- Secure login with email and password
- Role-based access control (User, Admin)

- Session management with secure tokens

Data Protection
- Secure API communication (HTTPS)
- Password hashing and encryption

System Controls
- Firewalls and network restrictions
- Controlled API access to AI services

Audit and Monitoring
- Logging of user activities (searches, bookmarks, feedback)
- Analytics for system monitoring

Backup and Recovery
- Backup storage on HDD
- Capability to restore system in case of failure

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

_____ Date: April 1, 2026

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